EVEREST
construction group

# ALUMINIUM PROFILES

## for doors, windows and facades

## Environmental Product Declaration

EPD
INTERNATIONAL EPD SYSTEM

**EPD OWNER:**
EVEREST

**PROGRAMME:**
THE INTERNATIONAL EPD SYSTEM, www.environdec.com

**PROGRAMME OPERATOR:**
EPD INTERNATIONAL AB

**TYPE OF EPD:**
EPD OF MULTIPLE PRODUCTS FROM A COMPANY,
BASED ON A REPRESENTATIVE PRODUCT: PROFILE 71 000
PROFILES ALSO INCLUDED IN THE EPD: 92 000, 93 000,
CE 32 000, E50, E34

**EPD REGISTRATION NUMBER:**
EPD-IES-0026335

**VERSION DATE:**
2025-10-08

**VALIDITY DATE:**
2030-10-07



In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021

An EPD may be depublished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com

ECO PLATFORM
EPD
VERIFIED

# GENERAL INFORMATION

## PROGRAMME INFORMATION

**PROGRAMME:** The International EPD System
**ADDRESS:** EPD INTERNATIONAL AB, box 21060, Se-100 31 Stockholm, Sweden
**WEBSITE:** www.environdec.com
**EMAIL:** support@environdec.com

## PRODUCT CATEGORY RULES (PCR)

**CEN STANDARD** EN 15804 serves as the Core Product Category Rules (PCR)

**PRODUCT CATEGORY RULES (PCR):** PCR 2019:14 Construction Products (EN 15804+A2) (VERSION 2.0.1), UN PC CODE 4212

**PCR REVIEW WAS CONDUCTED BY:** The Technical Committee Of The International Epd System. See www.environdec.com for a list of members. The review panel may be contacted via the secretariat www.environdec.com/contact.

## THIRD-PARTY VERIFICATION

Independent third-party verification of the declaration and data, according to ISO 14025:2006, via:

☑ **Individual EPD verification without a pre-verified LCA/EPD tool**

**Third-party verifier:** Luca Giacomello – Green&Care S.a.s.
**Approved by:** International EPD System

Procedure for follow-up of data during EPD validity involves third party verifier:

☐ YES ☑ NO

## LCA ACCOUNTABILITY

**COMPANY:** Life Cycle Engineering
**LCA PRACTITIONERS:** Davide Burlon, Stefano Donati
**CONTACT:** burlon@studiolce.it, donati@studiolce.it

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but published in different EPD programmes, may not be comparable. For two EPDs to be comparable, they shall be based on the same PCR (including the same first-digit version number) or be based on fuly aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical delcared/ functional units); have identical scope in terms of included life-cycle stages (unless the excluded life-cycle stage is demonstrated to be insignificant); apply identical impact assessment methods (including the same version of characterisation factors); and be valid at the time of comparison. For further information about comparability, see EN 15804 and ISO 14025.



# Information about the EPD owner

## THE EPD OWNER

**EPD OWNER:** EVEREST

**ADDRESS:** Rruga Bulevardi Blu, Nr. 544

**CONTACT:** Ermil Sita, ermil.sita@everest.al

**DESCRIPTION OF THE ORGANIZATION: Everest** is one of Europe's leading manufacturers of aluminium and hybrid profile systems, with over 25 years of experience in the industry. The company continues to expand its production capacity, driven by advanced technology, highly skilled personnel, and a strong commitment to innovation.

Everest designs and delivers tailored solutions for the construction and industrial sectors, offering aluminium and hybrid profiles for doors, windows, façades, and a wide range of applied systems.

With its proven reliability, superior product quality, and customer-oriented approach, Everest is recognized as a trusted partner across international markets.

**PRODUCT-RELATED OR MANAGEMENT SYSTEM-RELATED CERTIFICATIONS:** ISO 14001:2015



# PRODUCT INFORMATION

**REPRESENTATIVE PRODUCT:** PROFILE 71000

**OTHER PRODUCTS:** PROFILE 92000, PROFILE 93000, PROFILE CE 32000, PROFILE E 50, PROFILE E 34

## Product Description

**71 000 SERIES :** a **tilt-and-turn window system** that combines the **structural strength of aluminium** with **advanced thermal insulation performance**. With frame depths of **75–110 mm** and **Uf values up to 1.30 W/m²K**, the system is **certified according to EN 12207:2016**, **EN 12208:1999**, **and EN 12210:2016**. It balances **structural strength** with slim visible sections and offers both **standard and hidden frame variants**.

It is designed for use in **residential and commercial buildings**, offering **high energy efficiency**, **acoustic comfort**, and **durability**. The series includes both **standard sash** and **hidden sash**, providing flexibility in **performance** and **security applications**. It also offers the **option for outward opening**.

**Key features include:**

1. High thermal insulation with **34 mm polyamide barriers**.
2. **Design flexibility** with slim visible sections for **modern architecture**.
3. **Durability in varying climatic conditions** with **low maintenance requirements**.

**92000 SERIES** (**Lift & Slide**): Thermally insulated sliding system with large openings, **watertightness**, and **sound reduction**. Sash weight up to **400 kg**.

**93000 SERIES** (**Minimal Slide**): Ultra-slim sliding wall (visible frame from **10–20 mm**), floor-to-ceiling glazing, sash weight up to **500 kg**. Option for **concealed frame in awalls**.

**32000 SERIES** (**Railing Glass**): Glass railing system. **Easy interior installation**, compatible with glass thicknesses **66.4, 88.4**, **and 10.10.4**. Certified for **push force up to 300 kN**.

**E 50 SERIES** (**Curtain Wall**): Unitised facade with an **all-glass appearance**, Uw up to **0.7 W/m²K**, suitable for **high-rise projects**. The system also allows a **unitized version**.

**E 34 SERIES** (**Minimal Curtain Wall**): Slim **34 mm mask system** with **thermal break design**, certified according to **EN 13830**, for modern façades and thresholds for doors, of iron, steel or aluminium



PRODUCT INFORMATION

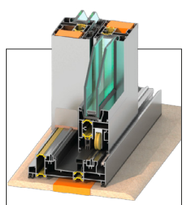


## 71000

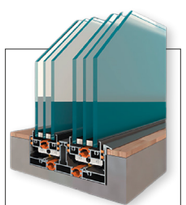
**Profile 71000** was chosen as representative product on the basis of production volumes and market sales.

**UN CPC code:** 4212 - Doors, windows and their frames

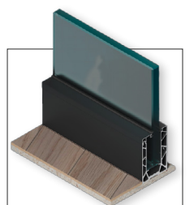
**Production site:** Rruga Bulevardi Blu, Nr.544
More information available at https://www.everest.al/



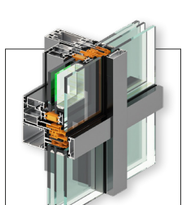
PROFILE 92000



PROFILE 93000



PROFILE CE 32000



PROFILE E 50



PROFILE E 34

# CONTENT DECLARATION

## Declared unit: 1kg

The content declaration indicates the composition of the reference product (71 000 series) and its packaging.

| PRODUCT CONTENT | MASS [KG] | POST-CONSUMER RECYCLED MATERIAL, MASS-% OF PRODUCT | PRE-CONSUMER RECYCLED MATERIAL, MASS-% OF PRODUCT | BIOGENIC MATERIAL, MASS-% OF PRODUCT | BIOGENIC MATERIAL, KG C/PRODUCT OR DECLARED UNIT |
|---|---|---|---|---|---|
| **ALUMINIUM** | 0.80 | 27 | 0 | 0 | 0 |
| **POLYAMIDE** | 0.14 | 0 | 0 | 0 | 0 |
| **PAINT** | 0.06 | 0 | 0 | 0 | 0 |
| **TOTAL** | 1.00 | 27 | 0 | 0 | 0 |

| PACKAGING MATERIALS | MASS [KG] | MASS-% (VERSUS THE PRODUCT) | BIOGENIC MATERIAL, KG C/PRODUCT OR DECLARED UNIT* |
|---|---|---|---|
| **PE (100% RECYCLED)** | 0.01 | 1.00 | 0 |
| **WOOD** | 0.02 | 1.90 | 0.009 |
| **CARDBOARD** | 0.003 | 0.30 | 0.001 |
| **POLYESTER** | 0.002 | 0.20 | 0 |

***No Substances of Very High Concern (SVHCs) are present in the product exceeding the 0.1% limit as set by the European Chemicals Agency***

**Biogenic carbon uptake and emission for packaging production in module A3 has been balanced.*





# LCA INFORMATION

**GEOGRAPHICAL SCOPE:** Albania, global

**DECLARED UNIT:** 1 kg of profile

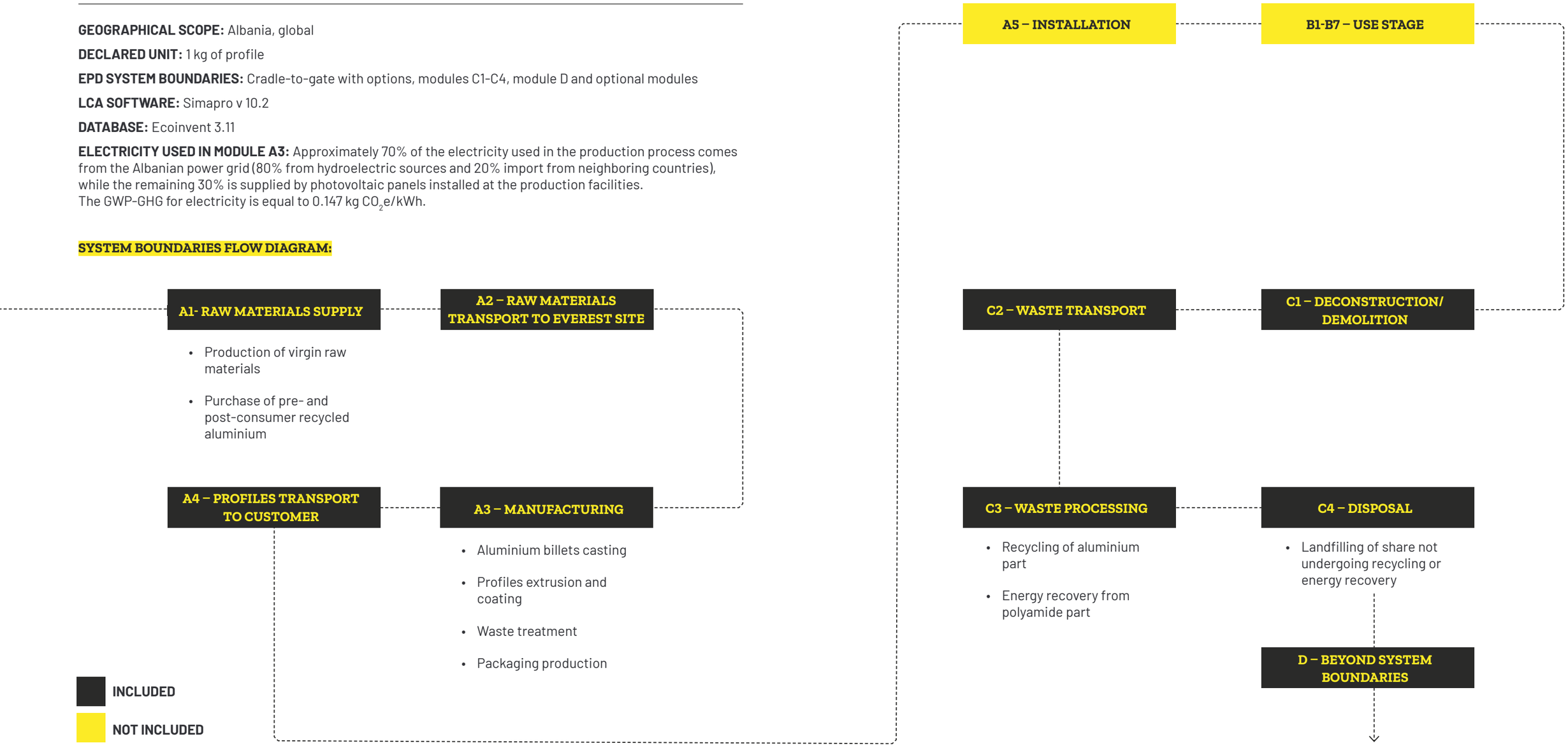
**EPD SYSTEM BOUNDARIES:** Cradle-to-gate with options, modules C1-C4, module D and optional modules

**LCA SOFTWARE:** Simapro v 10.2

**DATABASE:** Ecoinvent 3.11

**ELECTRICITY USED IN MODULE A3:** Approximately 70% of the electricity used in the production process comes from the Albanian power grid (80% from hydroelectric sources and 20% import from neighboring countries), while the remaining 30% is supplied by photovoltaic panels installed at the production facilities.
The GWP-GHG for electricity is equal to 0.147 kg $CO_2$e/kWh.

**SYSTEM BOUNDARIES FLOW DIAGRAM:**



# LCA INFORMATION

Modules declared, geographical scope, share of primary data (in GWP-GHG results) and data variation (in GWP-GHG results):

| | PRODUCT STAGE | | | CONSTRUCTION PROCESS STAGE | | USE STAGE | | | | | | | END OF LIFE STAGE | | | | BENEFITS AND LOADS BEYOND THE SYSTEM BOUNDARIES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Raw material supply | Transport | Manufacturing | Transport from the gate to the site | Assembly | Use | Maintenance | Repair | Replacement | Refurbishment | Operational energy use | Operational water use | De-construction demolition | Transport | Waste processing | Disposal | Reuse-Recovery-Recycling potential |
| **MODULE** | A1 | A2 | A3 | A4 | A5 | B1 | B2 | B3 | B4 | B5 | B6 | B7 | C1 | C2 | C3 | C4 | D |
| **MODULES DECLARED** | ✓ | ✓ | ✓ | ✓ | ND | ND | ND | ND | ND | ND | ND | ND | ✓ | ✓ | ✓ | ✓ | ✓ |
| **GEOGRAPHY** | GLO | GLO | AL | GLO | ND | ND | ND | ND | ND | ND | ND | ND | GLO | GLO | GLO | GLO | GLO |
| **PRIMARY DATA USED** | 25% | | | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| **VARIATION - PRODUCTS** | 19% | | | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| **VARIATION - SITES** | 0% | | | - | - | - | - | - | - | - | - | - | - | - | - | - | - |

✓ = module included in the analysis **ND** = module not included in the analysis **GLO** = Globe **AL** = Albania



# LCA INFORMATION

## Allocation procedures

The casting process, occurring in Everest plant, generates a **slag by-product** that is sold externally. An economic allocation was used to split the process natural gas consumption between the two outputs of the casting process: billets and the slag. The prices of the two different flows were provided directly by Everest. The economic allocation was performed considering also the production volumes of the two output products.

No economic allocation of flows was applied to distribute the impacts between the production of aluminium profiles and the waste generated by the same extrusion process, since discarded profiles are internally recycled in the casting process.

Post-consumer recycled aluminum is treated as burden-free, meaning that only the environmental impact associated with scrap processing and transportation to the production facility is accounted for. Internally recycled aluminum carries no associated impact, as it undergoes no additional treatment and is directly reintegrated into the manufacturing process.



# LCA INFORMATION

## Data quality assessment

The data quality assessment was carried out based on the PCR 2019:14 v2.0.1 and the EN 15941 standard "Data quality for environmental assessment of products and construction works". The parameters analysed include accuracy, completeness, and the degree of technological, geographical, and temporal coverage of all datasets that account for up to 80% of the environmental impact indicators.

Data on raw material consumption, processes, and transport to and from the factory come from primary sources, as they were obtained directly from the company. The degrees of geographical, temporal, and technological coverage were evaluated using the assessment table from the EN 15804:2012+A2:2019 standard. To obtain an overall evaluation for each of the three criteria analysed, a score from 1 to 5 was assigned, where 1 indicates "very poor" and 5 indicates "very good".

The evaluation of the three aspects is shown in the table below. Overall, the data quality was assessed as "Good", based on the information collected and the datasets used. Only one dataset received "poor" evaluation, the one adopted for the modelling of the LPG combustion, since the represented technology and geography are different from the ones under study, but no other dataset was available; the contribution of the dataset to the GWP-GHG is 21% and 9-23% for all other environmental impact indicators.

The share of primary data is calculated based on GWP-GHG results. It is a simplified indicator for data quality that supports the use of more primary data, to increase the representativeness of and comparability between EPDs. Note that the indicator does not capture all relevant aspects of data quality and is not comparable across product categories.

| GEOGRAPHICAL REPRESENTATIVENESS SCORE | TECHNOLOGICAL REPRESENTATIVENESS SCORE | TEMPORAL REPRESENTATIVENESS SCORE |
|---|---|---|
| 3.7 | 3.8 | 5.0 |

| PROCESS | SOURCE TYPE | SOURCE | REFERENCE YEAR | DATA CATEGORY | SHARE OF PRIMARY DATA, OF GWP-GHG RESULTS FOR A1-A3 |
|---|---|---|---|---|---|
| **RAW MATERIAL TRANSPORT** | Collected Data | EPD OWNER | 2024 | Primary Data | 1% |
| **ELECTRICITY CONSUMPTION FOR MANUFACTURING PROCESS** | Collected Data | EPD OWNER | 2024 | Primary Data | 2% |
| **GAS CONSUMPTION FOR MANUFACTURING PROCESS** | Collected Data | EPD OWNER | 2024 | Primary Data | 22% |
| **TOTAL SHARE OF PRIMARY DATA** | - | - | - | - | 25% |





# LCA INFORMATION

## Modelling of infrastructures

Power plants for the generation of electricity and casting furnace, both present in module A3, were included in the LCA model as relevant infrastructures.

No economic allocation of flows was applied to distribute the impacts between the production of aluminium profiles and the waste generated by the same extrusion process, since discarded profiles are internally recycled in the casting process.

## Downstream stages and module D

**C1 DECONSTRUCTION**

The demolition process is not directly managed by Everest; therefore, no primary data is available regarding the material flows required for dismantling the profiles.
As a result, a default value provided by PCR 2019:14 v2.0.1 was used, which indicates a diesel consumption of 0.0011 kWh per kg of dismantled material. The module includes both the production of diesel and the direct emissions resulting from its combustion.

**C2 WASTE TRANSPORT**

As this phase is also outside Everest's direct control, both the transport distance and the mode of transportation were taken from 2019:14 v2.0.1, which specifies a default distance of 80 km. The dataset used for land transport corresponds to Transport, freight, lorry 16-32 metric ton, EURO5 {RER}| transport, freight, lorry 16-32 metric ton, EURO5 | Cut-off, U.

**C3 WASTE TREATMENT**

A representative end-of-life scenario was derived from standard EN 17213 'Windows and doors – Environmental Product Declarations – Product category rules for windows and pedestrian doorsets' (March 2020). Although this standard is now expired, it was considered representative due to its specific focus on windows. The standard indicates that 95% of the product undergoes recycling/incineration with energy recovery. It was therefore assumed that 95% of the aluminium content and 95% of the polyamide (where present, depending on the composition type) are effectively recycled and incinerated with energy recovery, respectively. According to PCR 2019:14, default energy consumption values are assigned to the sorting phase, equal to 0.0018 kWh/kg of diesel and 0.0022 kWh/kg of electricity.

**C4 DISPOSAL**

As indicated by the EN 17213 standard, 5% by mass of the product is destined for disposal. Accordingly, 5% of the aluminium and polyamide content (where present), based on the different compositions, is assumed to be landfilled. As mentioned in the previous paragraph, all other raw materials (such as paint) are considered to be fully disposed of and are therefore entirely accounted for in Module C4:

It is realistically assumed that the paint follows the treatment of the aluminium, as it is adhered to it. Therefore, 95% of the paint is modeled as incinerated without energy recovery, based on the assumption that it degrades due to the high temperatures involved in remelting. The remaining 5% follows the profile to landfill and is modeled accordingly through landfill disposal operations.

Unlike Module C3, the environmental impacts of these treatments are fully allocated to the system under study.



**D BENEFITS AND LOADS BEYOND THE SYSTEM BOUNDARY**

The additional Module D allows for the assessment of potential benefits resulting from the recycling and energy recovery operations carried out at the product's end of life, as described in Module C3. These potential benefits are evaluated by comparing the end-of-life treatments with the processes they offset. In this study, the avoided processes are the production of primary aluminium, due to the use of post-consumer aluminium, and the generation of electricity and heat through the grid and natural gas combustion due to energy recovery from combustion of the polyamide bar, respectively. To assess the benefits from aluminium recycling, the amount of post-consumer recycled aluminium already present in the profiles at the end-of-life was excluded from the calculation to ensure a consistent mass balance. Moreover, only the benefit related to the primary aluminium content in the product was considered.

## Environmental impact assessment

The environmental impact was assessed using a dedicated method, in accordance with the EN 15804 standard and third-party verified and certified. The method is based on EF 3.1 package.



# LCA INFORMATION

## Environmental impact variations compared to reference product

The table below reports the indicators for which the results variation compared to the reference product is higher than 10%, considering the total of all modules A-C:

| IMPACT CATEGORY | INDICATOR | VAR % |
|---|---|---|
| **CLIMATE CHANGE-TOTAL** | GWP,t | 19% |
| **CLIMATE CHANGE-FOSSIL** | GWP,f | 19% |
| **CLIMATE CHANGE-LU&LU CHANGE** | GWP,luluc | 13% |
| **CLIMATE CHANGE - GWP GHG** | GWP - GHG | 19% |
| **ACIDIFICATION POTENTIAL** | AP | 15% |
| **EUTROPHICATION, FRESHWATER** | EPf | 63% |
| **EUTROPHICATION, MARINE** | EPm | 19% |
| **EUTROPHICATION, TERRESTRIAL** | EPt | 18% |
| **OZONE DEPLETION POTENTIAL** | ODP | 17% |
| **ADP FOR MINERALS AND METALS** | ADPe | 12% |
| **ADP FOR FOSSIL RESOURCES** | ADPf | 23% |
| **WATER USE** | WDP | 71% |

Such variations are all related to the product E 34, that differ from the reference product not having the polyamide thermal break and having the highest share of aluminium. Variations happen indeed more than 90% in the manufacturing stage A1-A3.




# ENVIRONMENTAL PERFORMANCE

The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3).

## Mandatory impact category indicators according to EN 15804

| IMPACT CATEGORY | UNIT | A1-3 | A4 | C1 | C2 | C3 | C4 | D | VAR % |
|---|---|---|---|---|---|---|---|---|---|
| **GWP,t** | kg $CO_2$ eq | 4.0E+00 | 1.1E-01 | 3.7E-04 | 1.2E-02 | 1.3E-03 | 1.4E-01 | -8.6E+00 | 19% |
| **GWP,f** | kg $CO_2$ eq | 4.0E+00 | 1.1E-01 | 3.7E-04 | 1.2E-02 | 1.3E-03 | 1.4E-01 | -8.6E+00 | 19% |
| **GWP,b** | kg $CO_2$ eq | 2.2E-03 | 3.4E-06 | 1.7E-08 | 3.9E-07 | 1.5E-06 | 2.5E-06 | -1.9E-03 | 7% |
| **GWP,luluc** | kg $CO_2$ eq | 4.3E-03 | 1.7E-06 | 1.5E-08 | 1.9E-07 | 2.2E-06 | 9.1E-07 | -1.2E-03 | 13% |
| **GWP - GHG** | kg $CO_2$ eq | 4.0E+00 | 1.1E-01 | 3.7E-04 | 1.2E-02 | 1.3E-03 | 1.4E-01 | -8.6E+00 | 19% |
| **AP** | mol H+ eq | 1.5E-02 | 2.9E-04 | 3.5E-06 | 3.2E-05 | 9.1E-06 | 1.5E-05 | -8.5E-02 | 15% |
| **EPf** | kg P eq | 9.8E-05 | 6.7E-08 | 3.5E-10 | 7.6E-09 | 7.1E-08 | 1.7E-08 | -6.0E-04 | 63% |
| **EPm** | kg N eq | 2.7E-03 | 1.1E-04 | 1.6E-06 | 1.2E-05 | 3.1E-06 | 7.2E-06 | -1.0E-02 | 19% |
| **EPt** | mol N eq | 2.9E-02 | 1.2E-03 | 1.8E-05 | 1.3E-04 | 3.4E-05 | 7.5E-05 | -1.1E-01 | 18% |
| **POCP** | kg NMVOC eq | 1.2E-02 | 4.7E-04 | 5.3E-06 | 5.2E-05 | 1.0E-05 | 2.0E-05 | -3.4E-02 | 7% |
| **ODP** | kg CFC11 eq | 1.1E-07 | 2.5E-09 | 5.7E-12 | 2.8E-10 | 2.0E-11 | 2.4E-11 | -2.6E-08 | 17% |
| **ADPe*** | kg Sb eq | 3.6E-06 | 2.8E-09 | 1.3E-11 | 3.2E-10 | 6.8E-11 | 9.1E-10 | 7.0E-07 | 12% |
| **ADPf*** | MJ | 6.0E+01 | 1.4E+00 | 4.9E-03 | 1.6E-01 | 2.5E-02 | 2.2E-02 | -9.0E+01 | 23% |
| **WDP*** | $m^3$ depriv. | 3.7E-01 | 4.7E-04 | 3.7E-06 | 5.3E-05 | 1.6E-04 | 0.0E+00 | 2.0E-01 | 71% |

**The results of this environmental impact indicator shall be used with care as the uncertainties of the results are high and as there is limited experience with the indicator.*

**GWP, t:** Global Warming Potential - total
**GWP,f:** Global Warming Potential - fossil
**GWP,b:** Global Warming Potential - biogenic
**GWP,luluc:** Global Warming Potential – land use and land use change
**GWP,GHG:** Global Warming Potential - irreversible
**ODP:** Ozone Depletion
**AP:** Acidification
**EPf:** Eutrophication - freshwater
**EPm:** Eutrophication - marine
**EPt:** Eutrophication - terrestrial
**POCP:** Photochemical Ozone Creation Potential
**ADPe:** Abiotic Depletion Potential - non-fossil
**ADPf:** Abiotic Depletion Potential - fossil
**WDP:** Water Deprivation Potential

# ENVIRONMENTAL PERFORMANCE

## Resource use indicators according to EN 15804

| IMPACT CATEGORY | UNIT | A1-3 | A4 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|---|
| **PERE** | MJ | 4.4E+00 | 3.6E-03 | 1.1E-05 | 4.0E-04 | 3.7E-03 | 3.7E-04 | -7.4E-01 |
| **PERM** | MJ | 3.4E-01 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 |
| **PERT** | MJ | 4.7E+00 | 3.6E-03 | 1.1E-05 | 4.0E-04 | 3.7E-03 | 3.7E-04 | -7.4E-01 |
| **PENRE** | MJ | 5.2E+01 | 1.4E+00 | 4.9E-03 | 1.6E-01 | 2.5E-02 | 2.2E-02 | -9.0E+01 |
| **PENRM** | MJ | 8.3E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | -1.6E+00 | 0.0E+00 | 0.0E+00 |
| **PENRT** | MJ | 6.0E+01 | 1.4E+00 | 4.9E-03 | 1.6E-01 | -1.5E+00 | 2.2E-02 | -9.0E+01 |
| **SM** | kg | 3.6E-01 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 4.3E-01 |
| **RSF** | MJ | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 |
| **NRSF** | MJ | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 |
| **FW** | $m^3$ | 1.9E-02 | 2.8E-05 | 1.5E-07 | 3.2E-06 | 1.0E-05 | 0.0E+00 | 2.4E-03 |

Method B reported in Annex 3 of the PCR is adopted for calculation of energy use indicators.

## Waste and output flows indicators according to EN 15804

| IMPACT CATEGORY | UNIT | A1-3 | A4 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|---|
| **HWD** | kg | 3.1E-01 | 2.6E-05 | 2.8E-07 | 2.9E-06 | 1.2E-05 | 1.9E-03 | -1.5E+00 |
| **NHWD** | kg | 1.6E+00 | 2.1E-03 | 4.8E-06 | 2.3E-04 | 7.7E-05 | 3.1E-01 | -3.8E-01 |
| **RWD** | kg | 3.5E-05 | 8.7E-08 | 2.3E-10 | 9.8E-09 | 1.2E-07 | 5.8E-09 | -1.1E-04 |
| **CRU** | kg | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 |
| **MFR** | kg | 3.7E-02 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 7.6E-01 | 0.0E+00 | 2.4E-02 |
| **MER** | kg | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 1.3E-01 | 0.0E+00 | 0.0E+00 |
| **EE** | MJ | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 0.0E+00 | 6.2E-01 | 0.0E+00 |

**PERE:** Renewable Primary Energy excluding Primary Energy used as raw material
**PERM:** Renewable Primary Energy used as raw material
**PERT:** Total use of Renewable Primary Energy
**PENRE:** Non-renewable Primary Energy excluding Primary Energy used as raw material
**PENRM:** Non-renewable Primary Energy used as raw material
**PENRT:** Total use of Non-renewable Primary Energy
**SM:** Use of secondary raw materials
**RSF:** Use of renewable secondary fuels
**NRSF:** Use of non-renewable secondary fuels
**FW:** Net use of fresh water
**HWD:** Hazardous waste disposed
**NHWD:** Non-hazardous waste disposed
**RWD:** Radioactive waste disposed
**CRU:** Components for re-use
**MFR:** Materials for recycling
**MER:** Materials for energy recovery
**EE:** Exported energy

# ADDITIONAL ENVIRONMENTAL INFORMATION

In accordance with the reference PCR, if the declared end-of-life scenarios correspond to a mix of end-of-life options, it is necessary to also report the results for the corresponding 100% scenarios. The results for modules C1–C4 related to environmental impact indicators are shown in the two tables below, for both the 100% recycling scenario—where aluminium is fully recycled and polyamide is entirely sent to energy recovery—and the 100% landfill scenario. In the 100% recycling scenario, module C4 still includes an impact contribution due to the incineration without energy recovery of the paint, as it melts together with the aluminium during the recycling process.

## 100% Landfill scenario

| IMPACT CATEGORY | UNIT | C1 | C2 | C3 | C4 |
|---|---|---|---|---|---|
| **GWP,t** | kg $CO_2$ eq | 3.7E-04 | 1.2E-02 | 0.0E+00 | 3.0E-02 |
| **GWP,f** | kg $CO_2$ eq | 3.7E-04 | 1.2E-02 | 0.0E+00 | 3.0E-02 |
| **GWP,b** | kg $CO_2$ eq | 1.7E-08 | 3.9E-07 | 0.0E+00 | 2.9E-05 |
| **GWP,luluc** | kg $CO_2$ eq | 1.5E-08 | 1.9E-07 | 0.0E+00 | 1.3E-05 |
| **GWP - GHG** | kg $CO_2$ eq | 3.7E-04 | 1.2E-02 | 0.0E+00 | 3.0E-02 |
| **AP** | kg CFC-11 eq | 3.5E-06 | 3.2E-05 | 0.0E+00 | 8.8E-05 |
| **EPf** | mol H+ eq | 3.5E-10 | 7.6E-09 | 0.0E+00 | 1.6E-07 |
| **EPm** | kg P eq | 1.6E-06 | 1.2E-05 | 0.0E+00 | 4.2E-05 |
| **EPt** | kg N eq | 1.8E-05 | 1.3E-04 | 0.0E+00 | 3.7E-04 |
| **POCP** | mol N eq | 5.3E-06 | 5.2E-05 | 0.0E+00 | 1.3E-04 |
| **ODP** | kg NMVOC eq | 5.7E-12 | 2.8E-10 | 0.0E+00 | 3.0E-10 |
| **ADPe** | kg Sb eq | 1.3E-11 | 3.2E-10 | 0.0E+00 | 2.7E-09 |
| **ADPf** | MJ | 4.9E-03 | 1.6E-01 | 0.0E+00 | 2.8E-01 |
| **WDP** | $m^3$ depriv. | 3.7E-06 | 5.3E-05 | 0.0E+00 | 0.0E+00 |

## 100% Recycling scenario

| IMPACT CATEGORY | UNIT | C1 | C2 | C3 | C4 |
|---|---|---|---|---|---|
| **GWP,t** | kg $CO_2$ eq | 3.7E-04 | 1.2E-02 | 1.3E-03 | 1.4E-01 |
| **GWP,f** | kg $CO_2$ eq | 3.7E-04 | 1.2E-02 | 1.3E-03 | 1.4E-01 |
| **GWP,b** | kg $CO_2$ eq | 1.7E-08 | 3.9E-07 | 1.5E-06 | 1.1E-06 |
| **GWP,luluc** | kg $CO_2$ eq | 1.5E-08 | 1.9E-07 | 2.2E-06 | 2.7E-07 |
| **GWP - GHG** | kg $CO_2$ eq | 3.7E-04 | 1.2E-02 | 1.3E-03 | 1.4E-01 |
| **AP** | kg CFC-11 eq | 3.5E-06 | 3.2E-05 | 9.1E-06 | 1.1E-05 |
| **EPf** | mol H+ eq | 3.5E-10 | 7.6E-09 | 7.1E-08 | 9.4E-09 |
| **EPm** | kg P eq | 1.6E-06 | 1.2E-05 | 3.1E-06 | 5.4E-06 |
| **EPt** | kg N eq | 1.8E-05 | 1.3E-04 | 3.4E-05 | 6.0E-05 |
| **POCP** | mol N eq | 5.3E-06 | 5.2E-05 | 1.0E-05 | 1.5E-05 |
| **ODP** | kg NMVOC eq | 5.7E-12 | 2.8E-10 | 2.0E-11 | 9.0E-12 |
| **ADPe** | kg Sb eq | 1.3E-11 | 3.2E-10 | 6.8E-11 | 8.1E-10 |
| **ADPf** | MJ | 4.9E-03 | 1.6E-01 | 2.5E-02 | 8.1E-03 |
| **WDP** | $m^3$ depriv. | 3.7E-06 | 5.3E-05 | 1.6E-04 | 8.4E-04 |

## Abbreviations

| STAGES / MODULES | |
|---|---|
| **A1** | Raw material supply |
| **A2** | Transport |
| **A3** | Manufacturing |
| **A4** | Transport to site |
| **A5** | Construction/Installation |
| **B1** | Use |
| **B2** | Maintenance |
| **B3** | Repair |
| **B4** | Replacement |
| **B5** | Refurbishment |
| **B6** | Operational energy use |
| **B7** | Operational water use |
| **C1** | Deconstruction/Demolition |
| **C2** | Transport to waste processing |
| **C3** | Waste processing |
| **C4** | Disposal |
| **D** | Reuse-Recovery-Recycling potential |

| OTHER RELEVANT TERMS | |
|---|---|
| **SVHC** | Substances of Very High Concern |
| **EC No.** | European Community Number |
| **CAS No.** | Chemical Abstracts Service Number |
| **MJ** | Megajoule |
| **kg** | Kilogram |
| **$m^3$** | Cubic Meter |
| **NMVOC** | Non-Methane Volatile Organic Compounds |
| **Sb eq.** | Antimony Equivalents |
| **P eq.** | Phosphorus Equivalents |
| **N eq.** | Nitrogen Equivalents |
| **CFC-11 eq.** | Chlorofluorocarbon-11 Equivalents |
| **$CO_2$ eq.** | Carbon Dioxide Equivalents |
| **kg C** | Kilograms of Carbon |
| **kg $CO_2$eq.** | Kilograms of Carbon Dioxide Equivalent |



# REFERENCES


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- EN 15804:2012+A2: Sustainability in construction. Product environmental statements. Commodity category rules for construction products.
- EN 17213 'Windows and doors – Environmental Product Declarations – Product category rules for windows and pedestrian doorsets' (March 2020)
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## Version History

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